

Systems with Analog Integrated Circuits

Analog Interface Design and Simulation

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Contents

1. Project Pillars	3
1.1. Requirement	3
1.2. Block diagram	3
1.3. Specifications	3
1.3.1. Stage 1: Instrumentation Amplifier (IA).....	3
1.3.2. Stage 2: Low-Pass Filter (Rauch Topology).....	4
1.3.3. Stage 3	4
1.3.4. Stage 4	5
2. Dimensioning	6
2.1. Stage 1	6
2.2. Stage 2	8
2.3. Stage 3	10
2.4. Stage 4	12
3. Circuit Stages Analysis and Characterization	12
3.1. Instrumentation amplifier (IA)	12
3.1.1. Operating Point.....	12
3.1.2. AC	13
3.1.3. Transient	17
3.2. Rauch Low-Pass Filter	19
3.2.1. Operating Point.....	19
3.2.2. AC	20
3.2.3. Transient	22
3.3. Noninverting Programmable Gain Amplifier with switches in the signal path	22
3.3.1. Operating Point.....	22
3.3.2. AC	23
3.3.3. Transient	25
3.3.4. Full-Wave Rectifier (FWR)	28
3.3.5. Operating Point.....	28
3.3.6. DC	28
3.3.7. Transient	30
4. Digital Interface for Analog Signal Control	32
5. Conclusion.....	35

1. Project Pillars

1.1. Requirement

The main goal of this project is to design an analog interface that connects a sensor output to an Analog-to-Digital Converter (ADC). My implementation consists of a signal conditioning chain divided into four main stages:

- Stage 1: Instrumentation Amplifier
- Stage 2: Active Low-Pass Filter (Rauch topology)
- Stage 3: Programmable Gain Amplifier (PGA)
- Stage 4: Precision Full-Wave Rectifier
-

1.2. Block diagram

To make the design easier to follow, I have first included a block diagram of the system below. In the following sections, I will describe each stage in detail, and the complete circuit schematic will be presented at the end of the report. The overall structure can be seen in Figure (1).

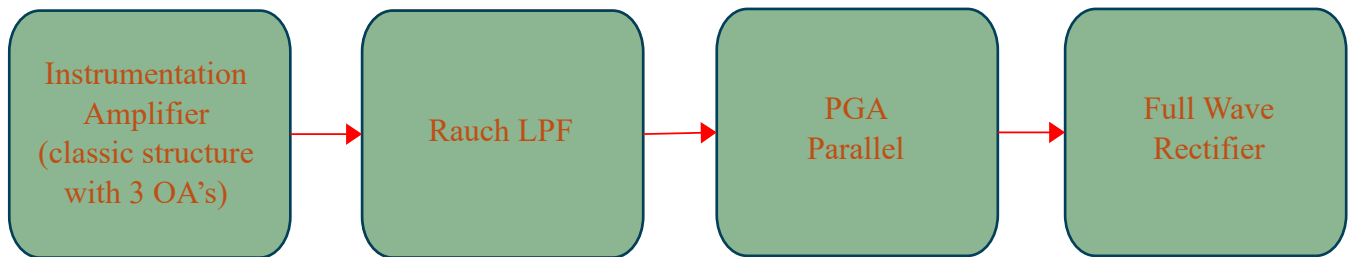


Figure (1): Block diagram

1.3. Specifications

The complete signal conditioning chain is designed to process a differential input signal through four distinct functional stages: amplification, filtering, gain control, and rectification.

1.3.1. Stage 1: Instrumentation Amplifier (IA)

The input source for the instrumentation amplifier is a differential voltage source (V-V) having a minimum amplitude of 44.1mV ($4.41 \cdot 10^{-2}\text{ V}$) and a maximum amplitude of 124mV ($1.24 \cdot 10^{-1}\text{ V}$). The circuit for the first stage is a classic three-operational amplifier instrumentation amplifier, chosen for its high common-mode rejection and high input impedance. The magnitude of the linear gain for this stage is 8 [V/V] (approx. 18.06 dB).

The circuit for the first stage is presented in figure (2).

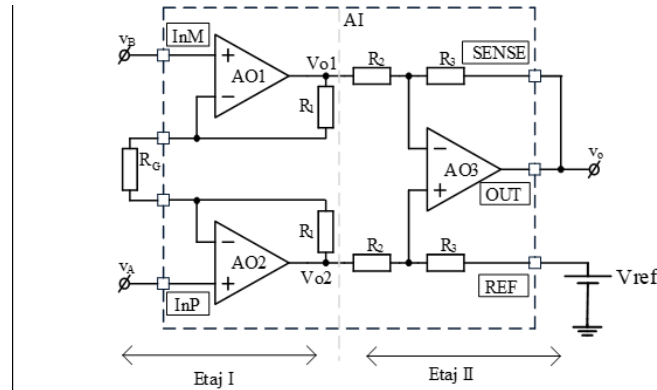


Figure (2): Instrumentation amplifier (3 OA's topology) with DC reference compensation

1.3.2. Stage 2: Low-Pass Filter (Rauch Topology)

The second stage of the signal-conditioning chain is a low-pass filter implemented using the Rauch (multiple-feedback) topology with a single operational amplifier, as shown in Figure (3).

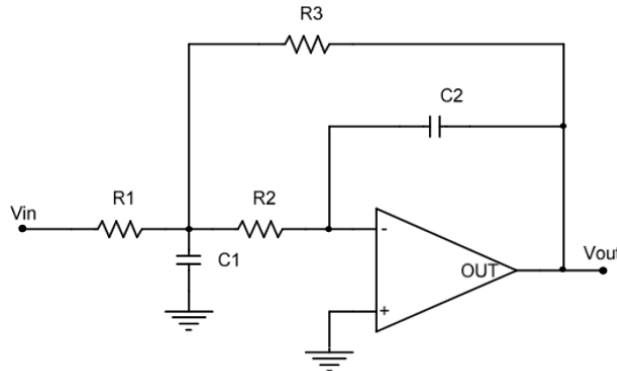


Figure (3): Rauch LPF

The minimum input resistance is $3 \text{ k}\Omega$, the bandwidth of the pass band is 1 kHz , and the quality factor is $Q = 1.41$. Moreover, the magnitude of the gain of this stage is 1 [V/V] in the pass band.

1.3.3. Stage 3

The Programmable Gain Amplifier (PGA) is implemented as a non-inverting operational amplifier with a parallel, digitally selectable feedback resistor network, as shown in Figure (4).

The PGA provides 4 discrete gain steps, with a minimum gain of 8 dB, a maximum gain of 17 dB, and a gain resolution of 3 dB per step: 8 dB, 11 dB, 14 dB and 17 dB.

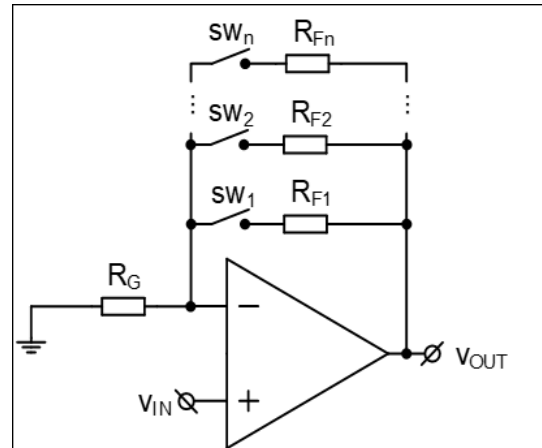


Figure (4): Programmable Gain Amplifier with switches in the signal path

The gain is digitally selected using a multiplexer that controls the feedback resistors; this implementation is discussed in a later chapter.

1.3.4. Stage 4

The full-wave rectifier is implemented using an operational-amplifier-based precision rectifier topology, as shown in Figure (5). This stage provides a linear voltage gain of 2 [V/V].

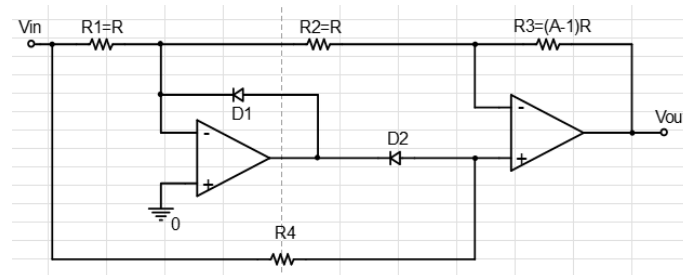


Figure (5): Full-wave-rectifier

2. Dimensioning

For the implementation, the stages were analyzed under ideal assumptions, as standard E24 resistor values were not used. The capacitors were also considered ideal.

2.1. Stage 1

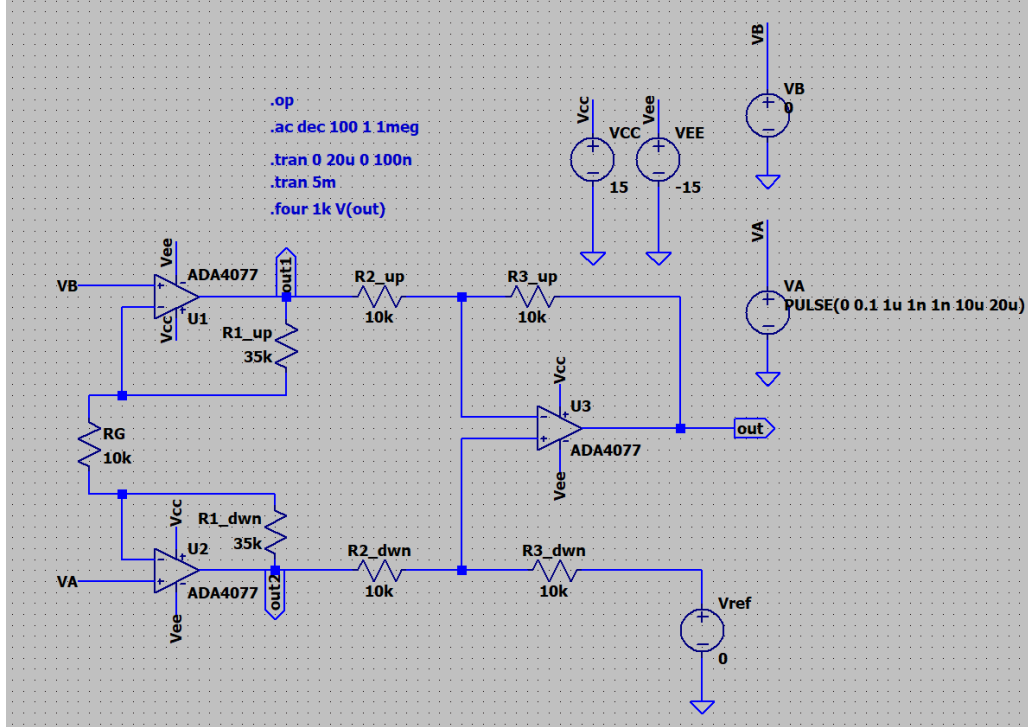


Figure (6): Implementation of 1st stage: IA

The first stage is designed as a three-op-amp instrumentation amplifier (IA) used to process the differential input signal. The design aims to achieve a total linear gain of 8 [V/V] (18.06 dB) while providing high input impedance to meet the minimum input resistance requirement of 3 k Ω .

The amplification process is divided into two distinct sections to maintain signal symmetry and maximize the Common-Mode Rejection Ratio (CMRR).

First Section Gain (A_{V1}): This section is comprised of two input buffers ($U1$ and $U2$). The gain is determined by the feedback resistors (R_{1_up} and R_{1_dwn}) and the gain-setting resistor (R_G).

The gain for this section is calculated using the formula:

$$A_{V1} = 1 + \frac{2 \cdot R_1}{R_G} \quad (1)$$

With $R_1 = 35k\Omega$ and $R_G = 10k\Omega$, the gain for this section is 8 [V/V].

Second Section Gain (A_{V2}): The second section consists of the differential amplifier ($U3$). To maintain balance and focus the primary amplification in the first section, this stage is configured for unitary gain (1 [V/V]). This is achieved by setting the differential resistors to equal values ($R_2=R_3=R_4=R_5=R_6 = 10k\Omega$).

$$A_{V2} = \frac{R_5}{R_3} = \frac{10k\Omega}{10k\Omega} = 1 \quad (2)$$

Total Gain (A_{total}): The final gain of the stage is the product of the gains from both sections:

$$A_{total} = A_{V1} \cdot A_{V2} = 8 \cdot 1 = 8 [V/V] \quad (3)$$

The components were selected to strictly adhere to the project's technical specifications:

- **Resistors:** The gain is set using $R_1 = 35k\Omega$ and $R_G = 10k\Omega$. The differential stage uses $10k\Omega$ resistors for R_2 through R_6 to ensure symmetry.
- **Operational Amplifiers:** The ADA4077 is utilized for all three positions ($U1$, $U2$, $U3$) due to its high precision and low offset characteristics.
- **Input Handling:** The stage is dimensioned to handle a minimum differential amplitude of 44.1 mV and a maximum amplitude of 124 mV.
- **Input Impedance:** Because the signal is applied to the non-inverting inputs of the ADA4077, the circuit naturally provides high input resistance, fulfilling the requirement for a minimum of 3 k Ω .
- **Reference:** The V_{ref} terminal is connected to 0V (GND) to provide ground-referenced output.

2.2. Stage 2

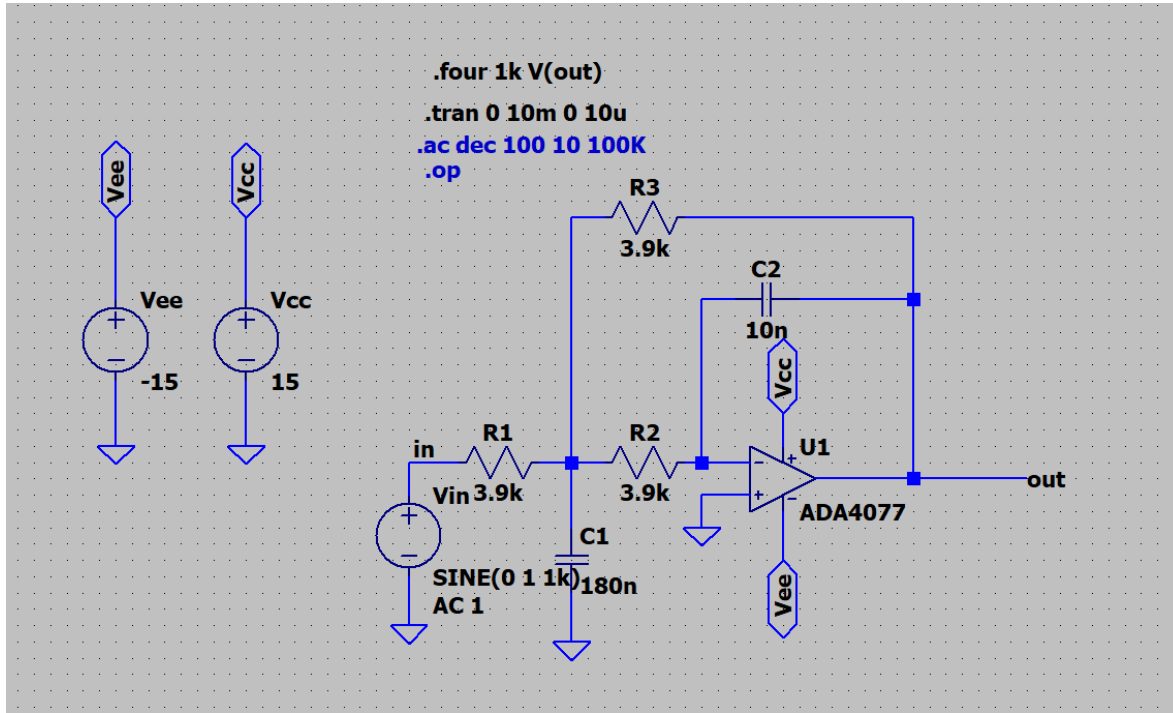


Figure (7): Implementation of the 2nd stage: Rauch LPF

The second stage of the system is implemented as an active second-order low-pass filter using the Rauch (multiple-feedback) topology, as shown in Figure (7). This stage is used to limit the signal bandwidth and to improve overall stability, while maintaining a linear behavior in the passband.

The filter is designed to have a unitary passband gain, $|H_0| = 1$ [V/V]. In the Rauch configuration, the passband gain is determined by the ratio of the feedback resistor to the input resistor, according to:

$$|H_0| = \frac{R_3}{R_1} \quad (4)$$

By choosing $R_1 = R_3 = 3.9 \text{ k}\Omega$, a unity gain is obtained:

$$|H_0| = \frac{3.9 \text{ k}\Omega}{3.9 \text{ k}\Omega} = 1 \text{ V/V} \quad (5)$$

The cutoff frequency of the filter is set to $f_c = 1$ kHz and is defined by the resistors R_2 , R_3 and the capacitors C_1 , C_2 as:

$$f_c = \frac{1}{2\pi\sqrt{R_2 R_3 C_1 C_2}} \quad (6)$$

Using $R_1 = R_2 = R_3 = 3.9$ k Ω , together with $C_1 = 180$ nF and $C_2 = 10$ nF, the resulting cutoff frequency satisfies the required bandwidth of 1 kHz.

The quality factor of the filter is specified as $Q = 1.41$. For the Rauch topology, the quality factor is given by:

$$Q = \frac{\sqrt{C_1/C_2}}{\sqrt{R_2/R_3} + \sqrt{R_3/R_2} + \sqrt{R_3/R_1}} \quad (7)$$

With equal resistor values, the expression simplifies to:

$$Q = \frac{\sqrt{C_1/C_2}}{3} \quad (8)$$

For the chosen capacitor ratio $C_1/C_2 = 18$, the resulting quality factor is approximately $Q = 1.41$, meeting the design requirement.

The selected resistor values also ensure a minimum input resistance of approximately $R_{in} \approx R_1 = 3.9$ k Ω , which is above the specified minimum of 3 k Ω and prevents loading of the previous stage. The operational amplifier used for this stage is the ADA4077, powered from a dual ± 15 V supply to guarantee sufficient linearity and dynamic range.

2.3. Stage 3

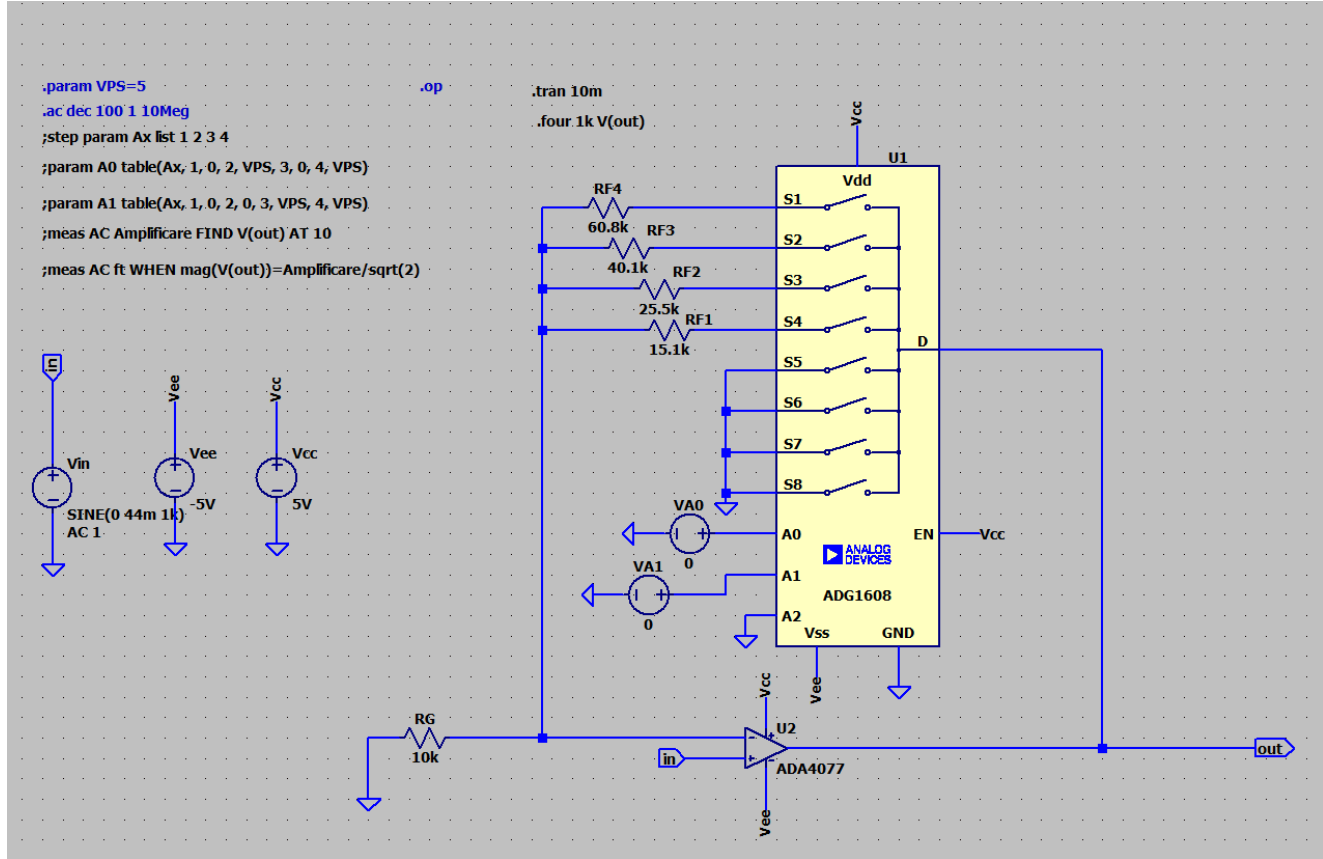


Figure (8): Implementation of the 3rd stage: parallel PGA

The programmable gain amplifier is built around a non-inverting operational amplifier ($U2$), where the gain is controlled by the feedback resistance selected through the ADG1608 analog multiplexer ($U1$). This approach allows digital control of the gain without introducing switches directly in the signal path. For a non-inverting amplifier, the voltage gain is given by:

$$A_v = 1 + \frac{R_F}{R_G} \quad (9)$$

In this design, the gain-setting resistor is fixed at $R_G = 10 \text{ k}\Omega$, while the feedback resistor R_F is selected from a resistor bank using the multiplexer. The feedback resistor values were calculated to provide a 3 dB increment between consecutive gain steps, starting from a minimum gain of 8 dB.

The corresponding gain values and resistor selections are:

- For **8 dB** ($A_v \approx 2.51$):

- $R_F = 15.1 \text{ k}\Omega$
- For **11 dB** ($A_v \approx 3.55$):
 $R_F = 25.5 \text{ k}\Omega$
- For **14 dB** ($A_v \approx 5.01$):
 $R_F = 40.1 \text{ k}\Omega$
- For **17 dB** ($A_v \approx 7.08$):
 $R_F = 60.8 \text{ k}\Omega$

As an example, for the minimum gain setting:

$$A_v = 1 + \frac{15.1 \text{ k}\Omega}{10 \text{ k}\Omega} = 2.51 \approx 8 \text{ dB} \quad (10)$$

Similarly, for the maximum gain setting:

$$A_v = 1 + \frac{60.8 \text{ k}\Omega}{10 \text{ k}\Omega} = 7.08 \approx 17 \text{ dB} \quad (11)$$

The gain selection is controlled digitally via the address lines $A0$ and $A1$, which command the ADG1608 multiplexer to connect the desired feedback resistor.

The ADA4077 operational amplifier was selected for this stage due to its high linearity and low noise, ensuring accurate amplification across all gain settings. The multiplexer operates from a dual $\pm 5 \text{ V}$ supply, while the operational amplifier is powered from the system's $\pm 15 \text{ V}$ rails, maintaining consistency with the previous stages.

Table (1) summarizes the digital control logic of the PGA stage, showing the correspondence between address lines, selected feedback resistors, and the resulting gain values.

A1	A0	Active MUX Channel	Selected Feedback Resistor	Linear Gain (A_v)	Gain [dB]
0	0	S1	RF4 = 60.8 k Ω	7.08	17 dB
0	1	S2	RF3 = 40.1 k Ω	5.01	14 dB
1	0	S3	RF2 = 25.5 k Ω	3.55	11 dB
1	1	S4	RF1 = 15.1 k Ω	2.51	8 dB

Table (1): Programmable Gain Amplifier (PGA) Logic Control and Gain Selection

Logic 0 = 0 V, logic 1 = 5 V on address lines $A1$ and $A0$

Gain is given by:

$$A_v = 1 + \frac{R_F}{R_G}, \quad R_G = 10 \text{ k}\Omega$$

2.4. Stage 4

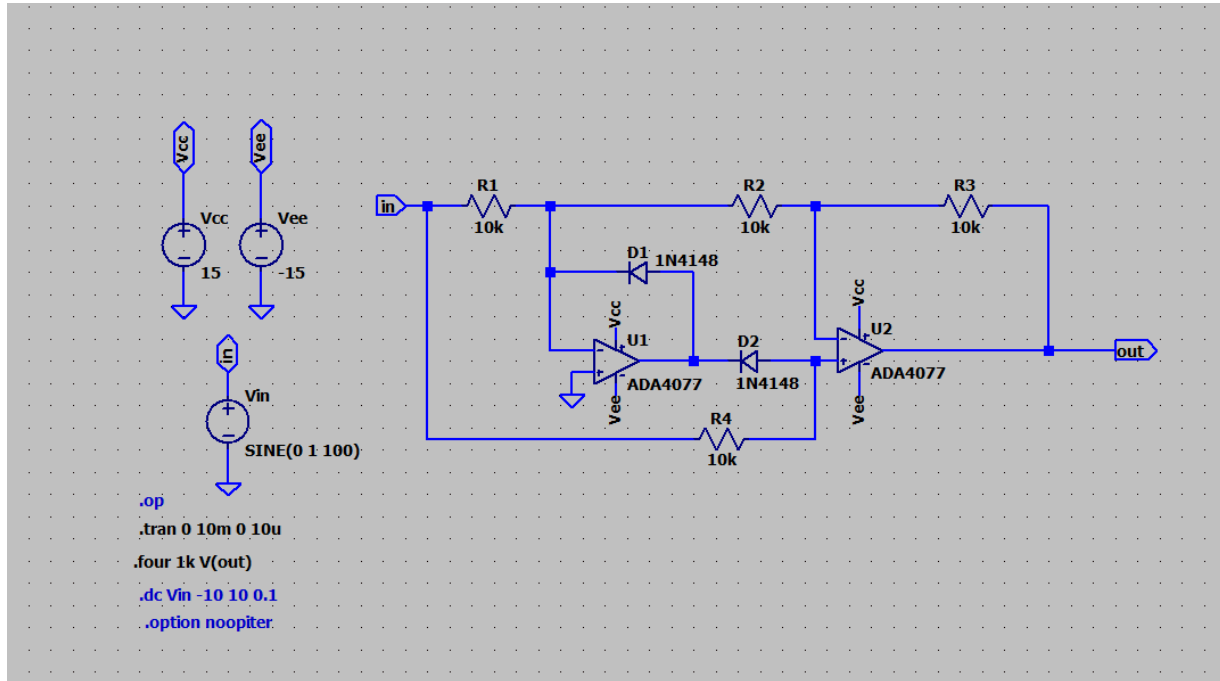


Figure (9): Implementation of the 4th stage: FWR

The final stage of the system is a precision full-wave rectifier, designed to convert the bipolar AC signal into a unipolar output suitable for DC measurement. The circuit provides a linear magnitude gain of 2 [V/V] and compensates the diode forward voltage using operational amplifiers, ensuring accurate rectification even for small signal amplitudes.

The stage is implemented using two ADA4077 operational amplifiers and two 1N4148 diodes, as shown in Figure (9). All resistors are chosen equal ($R_1 = R_2 = R_3 = R_4 = 10 \text{ k}\Omega$), resulting in the required gain and balanced operation. The circuit operates from a dual $\pm 15 \text{ V}$ supply.

3. Circuit Stages Analysis and Characterization

3.1. Instrumentation amplifier (IA)

3.1.1. Operating Point

An .op analysis was performed to verify the DC operating conditions of the instrumentation amplifier. The input sources were set to $V_A = 100 \text{ mV}$ and $V_B = 0 \text{ V}$, resulting in a differential input voltage of 100 mV.

The simulated output voltage is approximately 0.8 V, which corresponds to the expected gain of 8 [V/V]. All operational amplifiers operate in the linear region, and no saturation is observed.

--- Operating Point ---		
V(n001) :	0.225008	voltage
V(n002) :	1.02347e-05	voltage
V(n003) :	0.224998	voltage
V(n004) :	0.10001	voltage
V(n005) :	0	voltage
V(out) :	0.800018	voltage
V(out1) :	-0.350007	voltage
V(out2) :	0.449992	voltage
V(va) :	0.1	voltage
V(vb) :	0	voltage
V(vcc) :	15	voltage
V(vee) :	-15	voltage

Figure (10): Operating point of the 1st stage

3.1.2. AC

An AC analysis was performed to evaluate the frequency response of the instrumentation amplifier. The simulation shows a low-frequency gain of approximately 8 [V/V] (≈ 18 dB), in accordance with the specifications, with gain roll-off at higher frequencies due to the finite bandwidth of the operational amplifiers. See Figure (11):

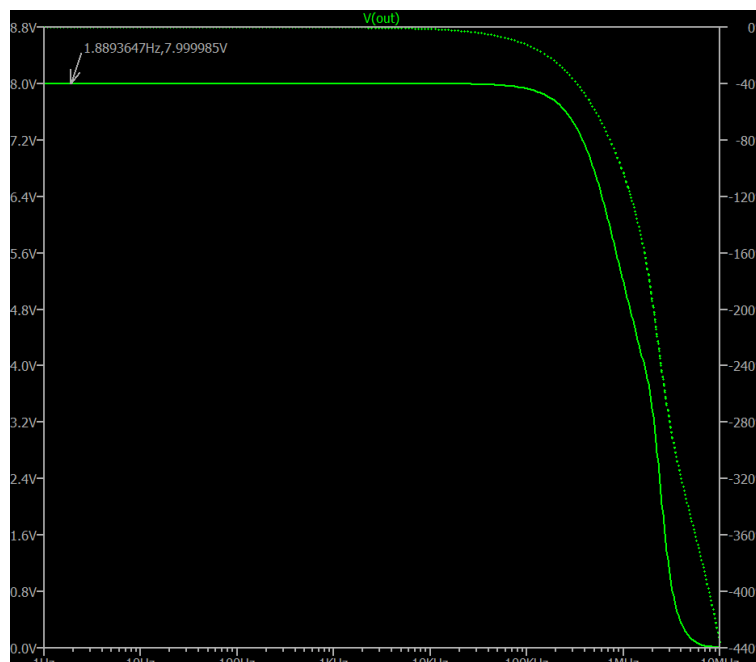


Figure (11): Gain at low frequency

The analysis “.ac dec 100 1 10meg” was used to study the frequency response of the first stage.

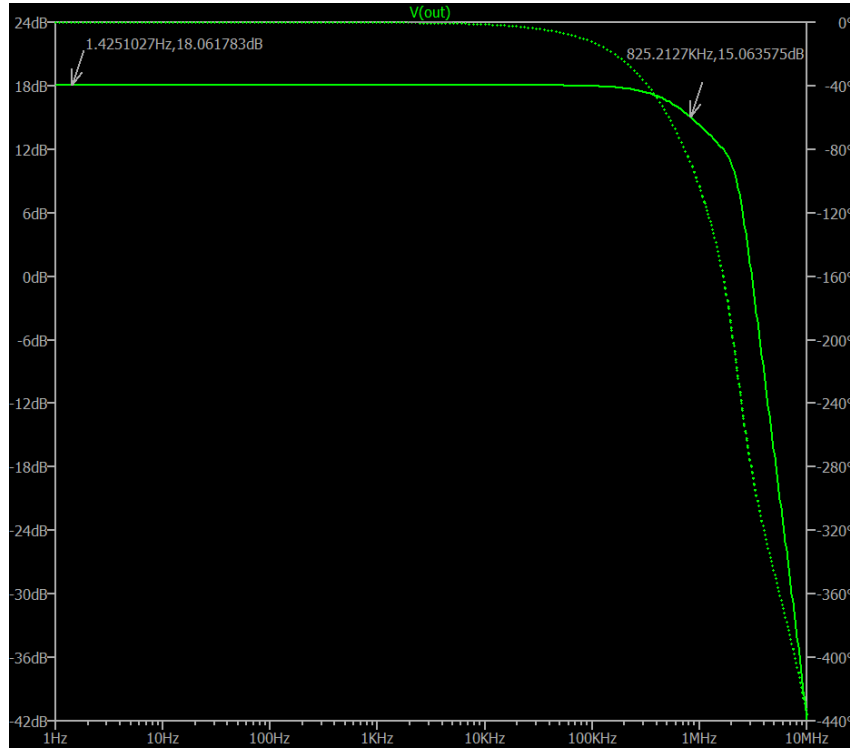


Figure (12): BW > Filter cutoff frequency (825kHz > 1kHz)

The common-mode rejection ratio (CMRR) is defined as the ratio between the differential gain (A_D) and the common-mode gain (A_{CM}):

$$CMRR = \frac{A_D}{A_{CM}} \quad (12)$$

In decibel scale, the CMRR is expressed as:

$$CMRR_{dB} = A_D(dB) - A_{CM}(dB) \quad (13)$$

From the AC simulations at low frequency, the differential gain is approximately 18.06 dB, while the common-mode gain is about -126 dB, resulting in a CMRR of approximately 144 dB, which confirms the excellent common-mode rejection of the instrumentation amplifier.

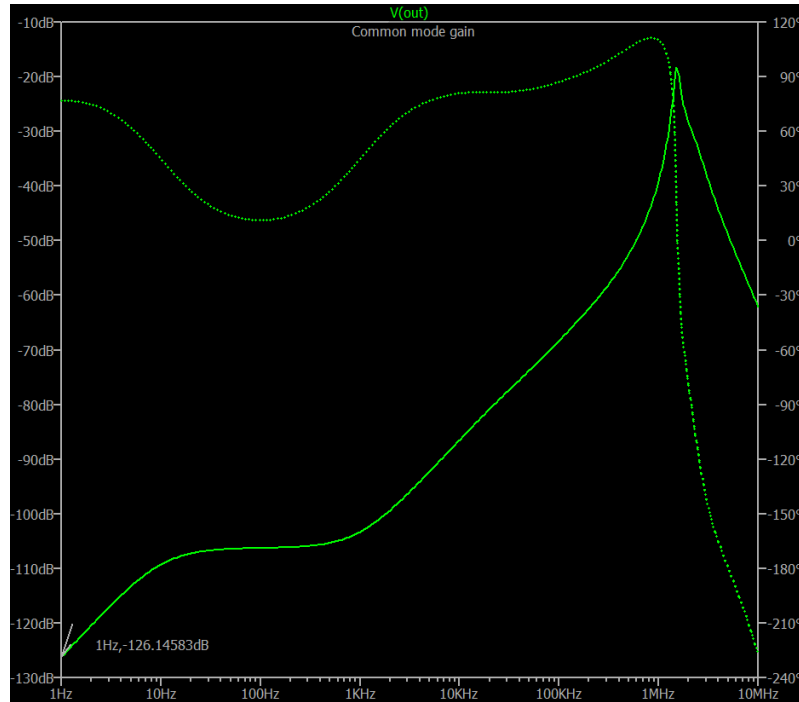


Figure (13): Common-mode Gain (ACM) at low freq. (-126dB)

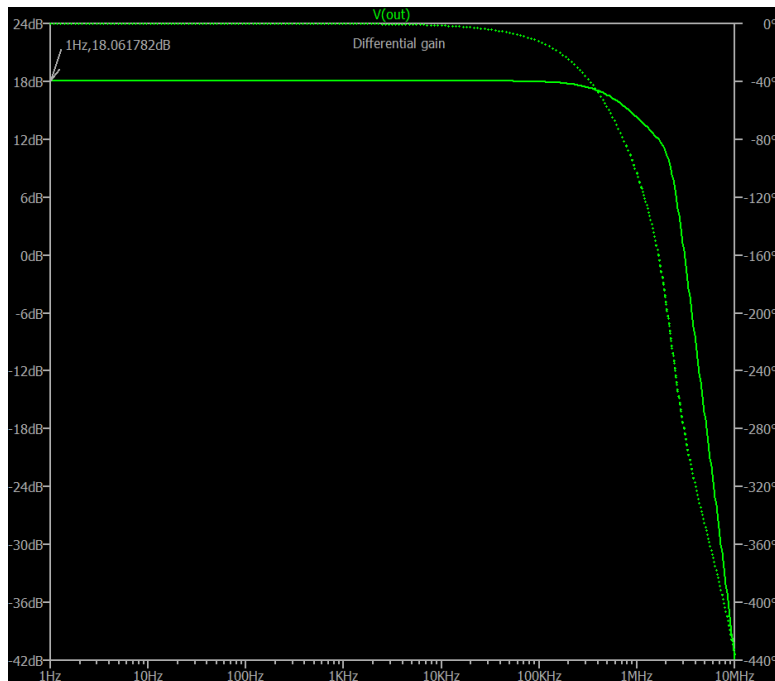


Figure (14): Differential Gain (AD) at low freq. (8dB)

The power supply rejection ratio (PSRR) quantifies the ability of the circuit to reject variations on the supply voltage. It is defined as the ratio between the differential gain and the gain from the power supply to the output:

$$\text{PSRR} = \frac{A_D}{A_{PS}} \quad (14)$$

In decibel scale, PSRR is expressed as:

$$\text{PSRR}_{\text{dB}} = A_D(\text{dB}) - A_{PS} \quad (15)$$

From the AC simulation, the power supply ripple gain at low frequency is approximately -119.6 dB at 1 Hz, while the differential gain is about 18.06 dB. This results in a PSRR of approximately 138 dB at low frequency, indicating excellent rejection of supply voltage variations.

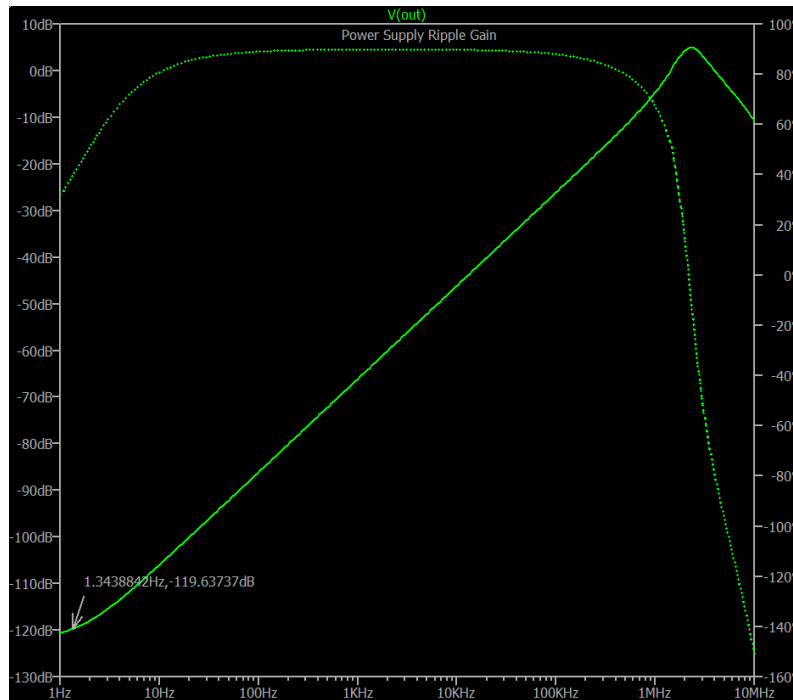


Figure (15): Power Supply Gain (APS)

3.1.3. Transient

A transient analysis (.tran 0 20u 100n) was performed to evaluate the slew rate of the instrumentation amplifier. A step signal was applied at the input using:

$V_A = \text{PULSE}(0, 0.1, 1 \mu s, 1 ns, 1 ns, 10 \mu s, 20 \mu s)$
And $V_B = 0$

From the simulation results, the output voltage changes by approximately 641 mV in 506 ns, resulting in a measured slew rate of:

$$SR \approx \frac{0.641 V}{0.506 \mu s} \approx \frac{1.27 V}{\mu s} \quad (16)$$

To verify that the measured Slew Rate of the instrumentation amplifier is adequate for the intended signal, it must be compared with the minimum Slew Rate required by the specifications.

For a sinusoidal output signal $v_{out}(t) = V_{pk} \sin(2\pi ft)$, the maximum slope is:

$$\left(\frac{dv_{out}}{dt} \right)_{\max} = 2\pi f \cdot V_{pk} \quad (17)$$

To avoid Slew-Rate-induced distortion, the following condition must be fulfilled:

$$\left(\frac{dv_{out}}{dt} \right)_{\max} \leq SR \Rightarrow 2\pi f \cdot V_{pk} \leq SR \quad (18)$$

In this case, the maximum frequency of interest is $f_{\max} = 1 \text{ kHz}$.

The maximum output peak amplitude is given by the maximum input amplitude multiplied by the IA low-frequency gain:

$$V_{out,pk,max} = A_v \cdot V_{in,pk,max} = 8 \cdot 0.124 V = 0.992 V \quad (19)$$

Thus, the required Slew Rate is:

$$SR_{req} = 2\pi \cdot 1000 \text{ Hz} \cdot 0.992 V = 6232 \text{ V/s} = 0.00623 \text{ V}/\mu \quad (20)$$

The Slew Rate measured from transient simulation is $SR_{meas} \approx 1.27 \text{ V}/\mu s$. Since:

$$SR_{req} = \frac{0.00623 V}{\mu s} \leq SR_{meas} = \frac{1.27 V}{\mu s} \quad (21)$$

So, the condition is satisfied, therefore the IA can reproduce the maximum expected output signal without Slew-Rate-related distortion.

This confirms that the amplifier can correctly follow fast input transitions within the intended operating range.

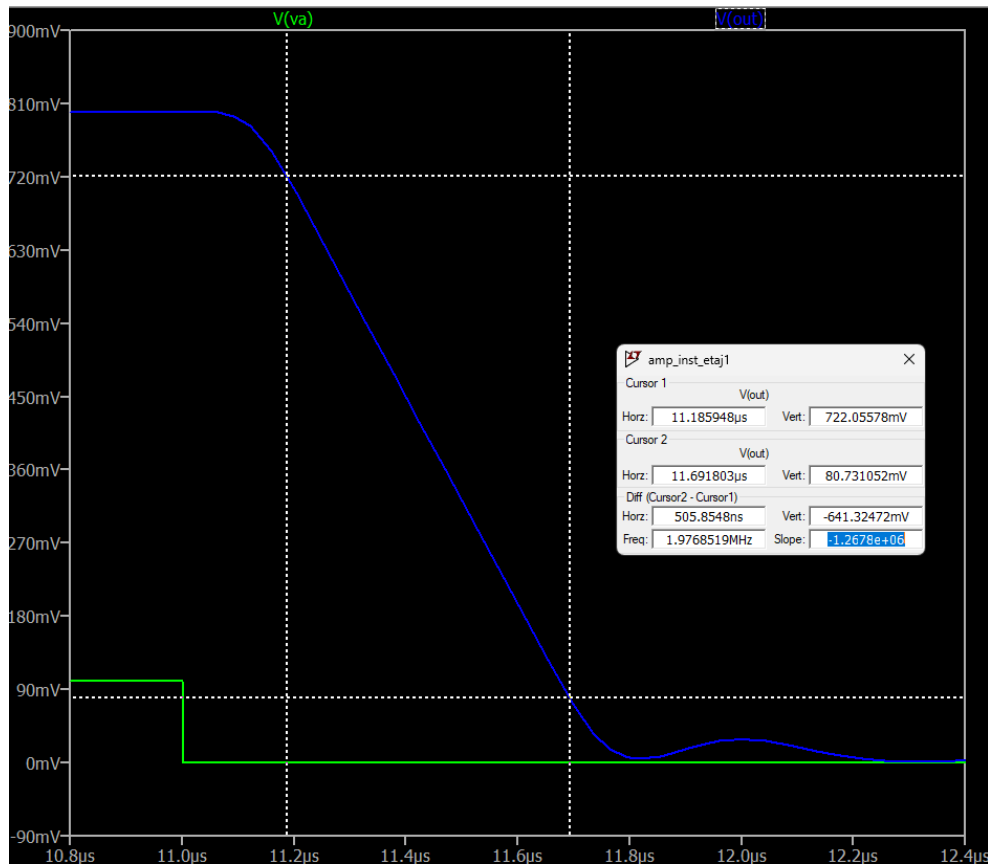


Figure (16): Testbench for measuring SR

A transient analysis (.tran 5m) followed by a Fourier analysis (.four 1k V(out)) was performed to evaluate the total harmonic distortion (THD) of the instrumentation amplifier. The input signal was applied as a sinusoid on one input, $V_B = \text{SINE}(0, 0.05, 1 \text{ kHz})$, while the other input was held at $V_A = 0 \text{ V}$.

The Fourier analysis shows a fundamental component at 1 kHz with negligible higher-order harmonics. The resulting total harmonic distortion is approximately 0.125%, confirming that the instrumentation amplifier exhibits low distortion and maintains linear operation for the specified input signal amplitude.

Fourier components of V(out)
N-Period=1
DC component: 2.65563e-05

Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [degree]
1	1.000e+03	4.000e-01	1.000e+00	-89.86°
2	2.000e+03	2.584e-05	6.461e-05	174.40°
3	3.000e+03	2.501e-04	6.254e-04	-134.07°
4	4.000e+03	4.938e-05	1.235e-04	-13.68°
5	5.000e+03	2.251e-04	5.627e-04	28.13°
6	6.000e+03	3.319e-05	8.297e-05	161.71°
7	7.000e+03	1.263e-04	3.157e-04	-95.78°
8	8.000e+03	4.697e-05	1.174e-04	168.00°
9	9.000e+03	2.675e-04	6.688e-04	146.07°

Partial Harmonic Distortion: 0.113793%
Total Harmonic Distortion: 0.125501%

Figure (17): THD for the 1st stage

3.2. Rauch Low-Pass Filter

3.2.1. Operating Point

An .op analysis was performed to verify the DC biasing of the Rauch low-pass filter and to check for output offset. The input source DC component was set to 1 V, and the simulated output settles close to 0 V ($V(out) \approx 1.0 \times 10^{-5}$ V), indicating negligible DC error and confirming that the stage operates in the linear region without saturation. The result of performing a “.op” analysis is illustrated in figure (18).

```

--- Operating Point ---
V(in):      1          voltage
V(n001):    2.56415e-07 voltage
V(n002):    1.01535e-05 voltage
V(out):     1.01535e-05 voltage
V(vcc):     15         voltage
V(vee):     -15        voltage
I(C1):      0.000256415 device_current
I(C2):      2.03885e-09 device_current
I(R1):      -0.00025641 device_current
I(R2):      2.53772e-09 device_current
I(R3):      2.53772e-09 device_current
I(Vcc):     -0.000400022 device_current
I(Vee):     0.000400022 device_current
I(Vin):     -0.00025641 device_current
Ix(u1:1):   -4.01135e-10 subckt_current
Ix(u1:2):   -4.98865e-10 subckt_current
Ix(u1:3):   0.000400021 subckt_current
Ix(u1:4):   -0.000400021 subckt_current
Ix(u1:5):   -4.57657e-09 subckt_current

```

Figure (18): Operating Point of the 2nd stage

3.2.2. AC

A “.ac dec 100 10 100k” analysis was performed to determine the cutoff frequency f_c , the pass-band gain H_0 , and to verify the frequency response of the Rauch low-pass filter. The simulation confirms a pass-band gain of approximately 1 [V/V] (0 dB) and a cutoff frequency of about 1 kHz, in accordance with the specifications.

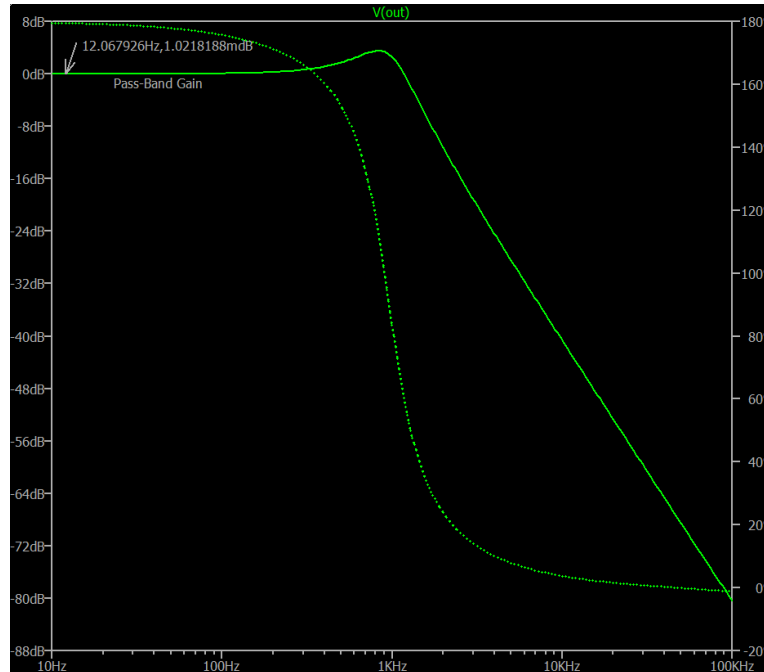


Figure (19): Analysis of the LPF

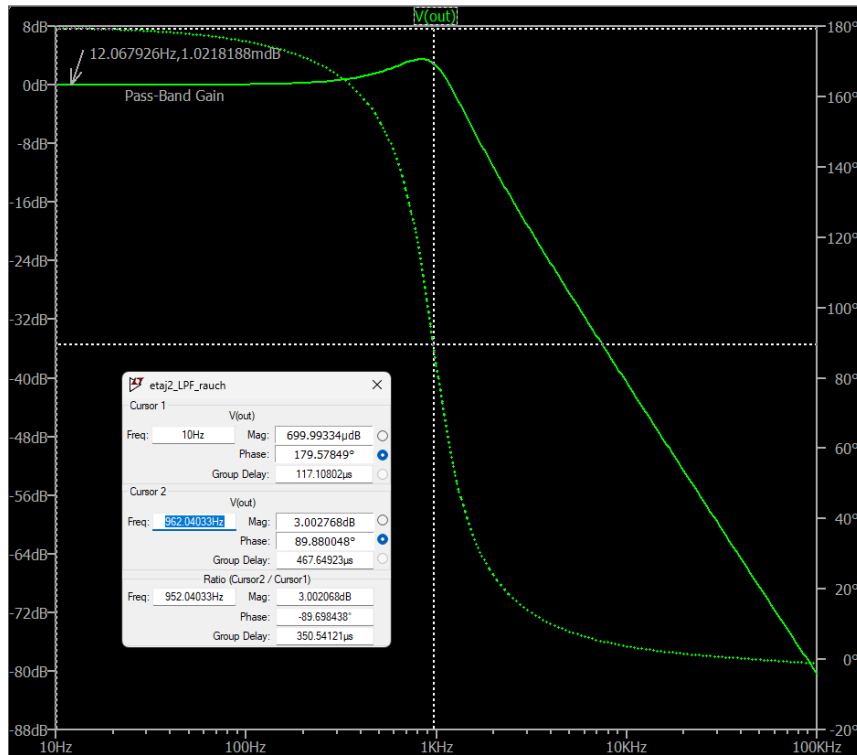


Figure (20): BW of the Filter

The bandwidth of the Rauch low-pass filter was determined from the phase response, by identifying the frequency at which the phase reaches -90° , which occurs at approximately 960 Hz, in accordance with the specified bandwidth of 1 kHz.

	f_c [Hz]	BW [Hz]	Q	H_0 [V/V]
Measured	960	960	1.41	1.002
Theoretical	1000	1000	1.41	1.000
Relative Error	4%	4%	0%	0.2%

Table (2): LPF measured vs theoretical values comparison

For the Rauch low-pass filter, the bandwidth is equal to the cutoff frequency, $BW = f_c$. The quality factor Q cannot be directly extracted from the magnitude response, since for $Q > 0.707$ the low-pass filter exhibits a **gain peak (overshoot)** around the cutoff frequency. Instead of having a well-defined -3 dB point, the response shows a local increase in gain near f_c , which makes the classical -3 dB method inapplicable. Therefore, Q is treated as a design parameter, taken from the specifications and calculated analytically based on the component values. For the chosen configuration with $R_1 = R_2 = R_3 = R$, the

quality factor is given by:

$$Q = \frac{1}{3} \cdot \sqrt{\frac{C_1}{C_2}} = \frac{1}{3} \cdot \sqrt{\frac{180nF}{10nF}} \approx 1.41 \quad (22)$$

3.2.3. Transient

The same analysis as in Section 3.1.3 was used to determine the total harmonic distortion (THD) of the second stage. A transient simulation followed by a Fourier analysis (.four 1k V(out)) was performed for an input signal $V_{in} = \text{SINE}(0, 1, 1 \text{ kHz})$. The resulting THD is approximately 0.043%, confirming linear operation of the Rauch low-pass filter for the specified input conditions.

Fourier components of V(out)
N-Period=1
DC component:3.99079e-05

Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [degree]	Normalized Phase [deg]
1	1.000e+03	1.351e+00	1.000e+00	6.37Å°	0.00Å°
2	2.000e+03	2.117e-05	1.567e-05	-89.91Å°	-96.28Å°
3	3.000e+03	1.435e-04	1.062e-04	137.87Å°	131.50Å°
4	4.000e+03	4.074e-05	3.016e-05	-18.98Å°	-25.34Å°
5	5.000e+03	9.729e-05	7.202e-05	-23.36Å°	-29.73Å°
6	6.000e+03	2.872e-05	2.126e-05	-24.68Å°	-31.05Å°
7	7.000e+03	3.334e-05	2.468e-05	76.25Å°	69.88Å°
8	8.000e+03	8.785e-05	6.503e-05	-126.03Å°	-132.40Å°
9	9.000e+03	2.138e-04	1.582e-04	112.33Å°	105.96Å°

Partial Harmonic Distortion: 0.021897%
Total Harmonic Distortion: 0.043007%

Figure (21): THD for the 2nd stage

3.3. Noninverting Programmable Gain Amplifier with switches in the signal path

3.3.1. Operating Point

For the operating point analysis of the programmable gain amplifier, the input was set to $V_{in} = 1 \text{ V DC}$ in order to verify the DC gain of the selected configuration. The simulation shows an output voltage of approximately $V_{out} = 4.09 \text{ V}$, corresponding to a linear gain of about 4.1 V/V, which is consistent with the selected feedback resistor of the PGA. All internal node voltages settle to stable values and the output remains within the $\pm 5 \text{ V}$ supply rails, confirming correct biasing and linear operation of the stage.

The result of performing a “.op” analysis is illustrated in figure (22).

```

--- Operating Point ---

V(in) :          1          voltage
V(n001) :        0.635299   voltage
V(n002) :        4.08558    voltage
V(n003) :        0.635249   voltage
V(n004) :        0.635267   voltage
V(n005) :        0.63528    voltage
V(n006) :        0          voltage
V(n007) :        0          voltage
V(out) :         4.08581     voltage
V(vcc) :         5          voltage
V(vee) :        -5          voltage

```

Figure (22): Operating Point of 3rd stage

3.3.2. AC

In order to observe all gain steps simultaneously, two voltage sources that control the address bits A0 and A1 of the multiplexer were used. The DC values of these sources are {A0} and {A1}, which are parameters obtained from lookup tables controlled by another parameter, Ax. For example, when the parameter Ax takes the third value from the list, $Ax = 3$, the parameter A0 becomes 0 V and the parameter A1 becomes 5 V. A value of 0 V corresponds to logic 0, while a value of 5 V corresponds to logic 1.

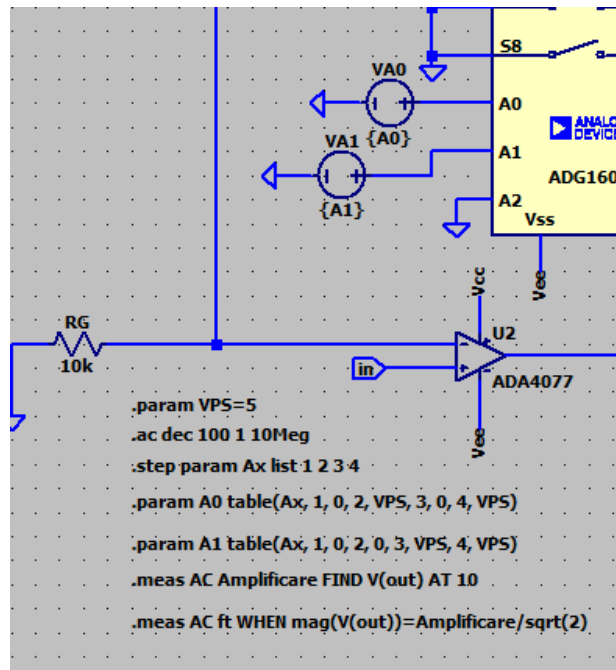


Figure (23): Configuring the testbench for AC analysis of 3rd stage

By this method, in each step a single internal switch of the multiplexer is closed, as presented in Table (1) in Section 2.3.

The gain corresponding to each programmable step is annotated directly on the AC magnitude response shown in Figure (24). Measuring the bandwidth for each gain configuration was not necessary, since the cutoff frequencies are clearly higher than the maximum passband frequency of the previous stage, as observed from the horizontal axis. A comparison between the required gain values and the measured results is presented in the following table:

	A_{vv1} [dB]	A_{vv2} [dB]	A_{vv3} [dB]	A_{vv4} [dB]
Measured	7.995	11.0057	13.9977	17.0015
Required	8	11	14	17
Relative error	0.062%	0.052%	0.016%	0.009%

Table (3): PGA measured vs. required values comparison

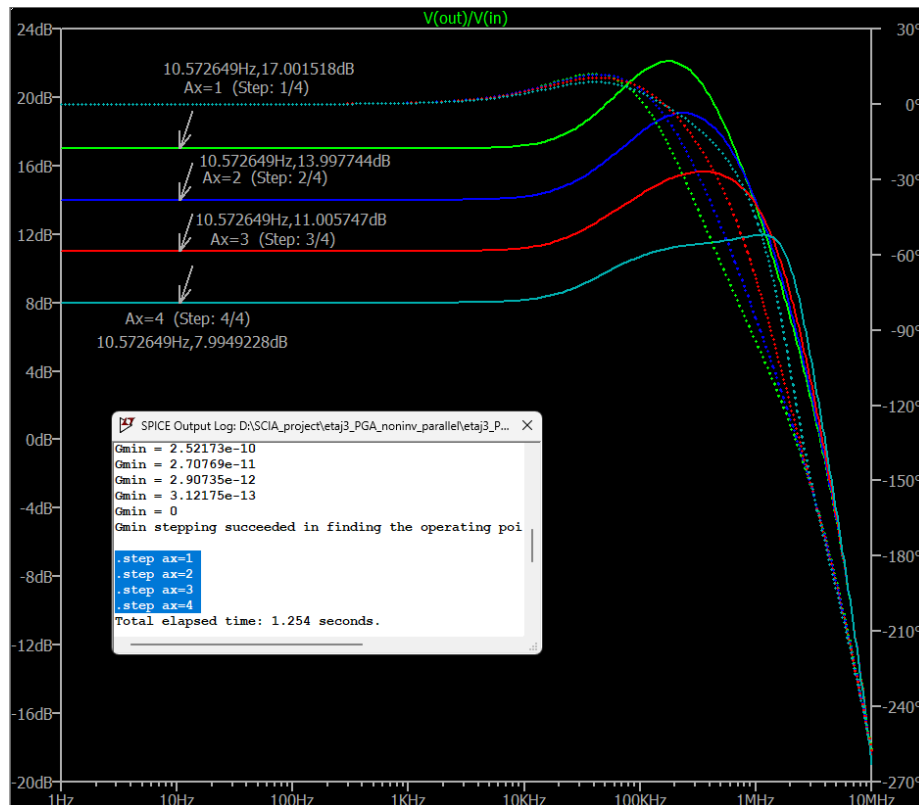


Figure (24): Low-frequency gain for each branch of the PGA

3.3.3. Transient

For this stage, the total harmonic distortion (THD) was evaluated in two operating conditions. In the first case, the input amplitude of the first stage is minimum and the PGA is set to its maximum gain (17 dB) by applying $A_0 = A_1 = 0$ V (Figure 25). In the second case, the input amplitude of the first stage is maximum and the PGA is set to its minimum gain (8 dB) by applying $A_0 = A_1 = 5$ V (Figure 24). In both cases, a transient analysis (.tran 10m) followed by a Fourier analysis (.four 1k V(out)) was performed.

Since the transient plot is not expressed in dB, the gain was evaluated from the time-domain amplitudes using:

$$A_v = \frac{V_{out}}{V_{in}} \quad (23)$$

For the minimum-gain configuration, the measured amplitudes are $V_{out} \approx 310$ mV and $V_{in} \approx 123$ mV, resulting in $A_v \approx 2.52$, corresponding to approximately 8 dB. The Fourier analysis indicates a THD of about 0.099% for this operating point.

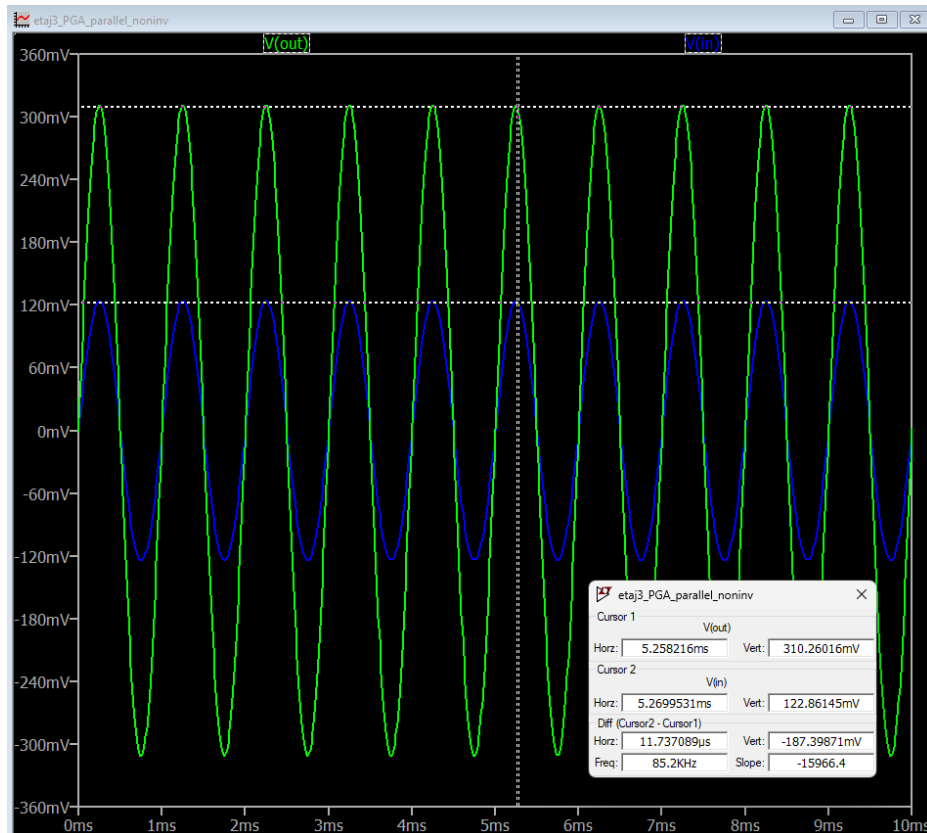


Figure (25): Transient Response of the PGA at Minimum Gain (8 dB)
Used for THD Calculation

Fourier components of V(out)
N-Period=1
DC component:-8.35859e-07

Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [degree]	Normalized Phase [deg]
1	1.000e+03	3.112e-01	1.000e+00	89.54°	0.00°
2	2.000e+03	3.440e-06	1.105e-05	151.60°	62.06°
3	3.000e+03	1.939e-04	6.232e-04	103.35°	13.81°
4	4.000e+03	4.911e-06	1.578e-05	-51.38°	-140.92°
5	5.000e+03	1.107e-04	3.556e-04	-61.85°	-151.39°
6	6.000e+03	4.310e-06	1.385e-05	119.83°	30.29°
7	7.000e+03	1.135e-05	3.646e-05	39.67°	-49.87°
8	8.000e+03	2.188e-06	7.031e-06	-168.62°	-258.16°
9	9.000e+03	1.201e-04	3.859e-04	131.70°	42.16°

Partial Harmonic Distortion: 0.081585%
Total Harmonic Distortion: 0.099385%

Figure (26): THD for minimum PGA gain

For the maximum-gain configuration, the PGA was set to its highest gain by selecting logic 0–0 on the address lines, while the input signal amplitude was reduced to avoid saturation. The measured time-domain amplitudes are $V_{out} \approx 311$ mV and $V_{in} \approx 44$ mV, resulting in a voltage gain of:

$$A_v = \frac{V_{out}}{V_{in}} \approx \frac{311 \text{ mV}}{44 \text{ mV}} \approx 7.1 \quad (24)$$

Which corresponds to approximately 17 dB, in agreement with the design specification. The Fourier analysis of the output signal indicates a Total Harmonic Distortion (THD) of approximately 0.109%, confirming that the amplifier maintains low distortion even at maximum gain.

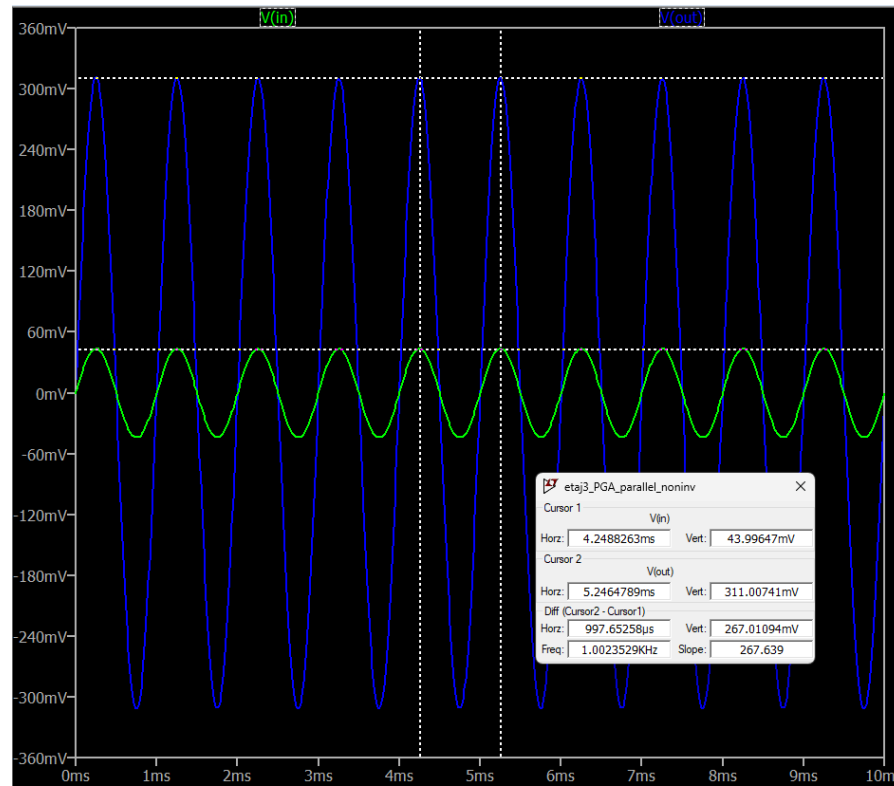


Figure (27): Transient Response of the PGA at Maximum Gain (17 dB)
Used for THD Calculation

Fourier components of V(out)
N-Period=1
DC component:-1.54867e-05

Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [degree]	Normalized Phase [deg]
1	1.000e+03	3.115e-01	1.000e+00	89.43°	0.00°
2	2.000e+03	4.430e-06	1.422e-05	135.29°	45.86°
3	3.000e+03	2.115e-04	6.790e-04	110.36°	20.93°
4	4.000e+03	7.116e-06	2.285e-05	-64.44°	-153.87°
5	5.000e+03	1.123e-04	3.606e-04	-58.76°	-148.20°
6	6.000e+03	6.102e-06	1.959e-05	125.50°	36.07°
7	7.000e+03	4.690e-05	1.506e-04	-12.65°	-102.08°
8	8.000e+03	1.005e-05	3.228e-05	154.08°	64.65°
9	9.000e+03	1.431e-04	4.593e-04	139.38°	49.95°

Partial Harmonic Distortion: 0.090935%
Total Harmonic Distortion: 0.108715%

Figure (28): THD for maximum PGA gain

3.3.4. Full-Wave Rectifier (FWR)

3.3.5. Operating Point

The result of performing a “.op” analysis is illustrated in figure (28).

```

--- Operating Point ---
V(n001) :      1.0478e-05      voltage
V(in) :        1              voltage
V(n002) :     -0.999959      voltage
V(out) :      -1.99947      voltage
V(n003) :     -1.51132      voltage
V(vcc) :       15            voltage
V(vee) :      -15            voltage
V(n004) :     -0.999969      voltage
I(D1) :      -2.52151e-09     device_current
I(D2) :       0.000199997     device_current
I(R1) :      -9.99989e-05     device_current
I(R2) :      -9.99969e-05     device_current
I(R3) :      -9.99511e-05     device_current
I(R4) :      -0.000199997     device_current
I(Vcc) :     -0.000500079     device_current
I(Vee) :       0.000500067     device_current
I(Vin) :     -0.000299996     device_current
Ix(u1:1) :    -4.01171e-10     subckt_current
Ix(u1:2) :    -4.98829e-10     subckt_current
Ix(u1:3) :     0.000200014     subckt_current
Ix(u1:4) :    -0.000200011     subckt_current
Ix(u1:5) :     0.0002         subckt_current
Ix(u2:1) :    -4.15495e-10     subckt_current
Ix(u2:2) :    -5.13111e-10     subckt_current
Ix(u2:3) :     0.000300062     subckt_current
Ix(u2:4) :    -0.000300058     subckt_current
Ix(u2:5) :     9.99511e-05     subckt_current

```

Figure (29): Operating Point of 4th stage

3.3.6. DC

The gain can be determined by the slope of the linear region of the voltage transfer characteristic (VTC). The input voltage swing can be found as well. The full bridge rectifier behavior can be seen, as the slope is positive for negative values of the input voltage and it is negative, respectively, for positive values of the input signal. See Figure (30):

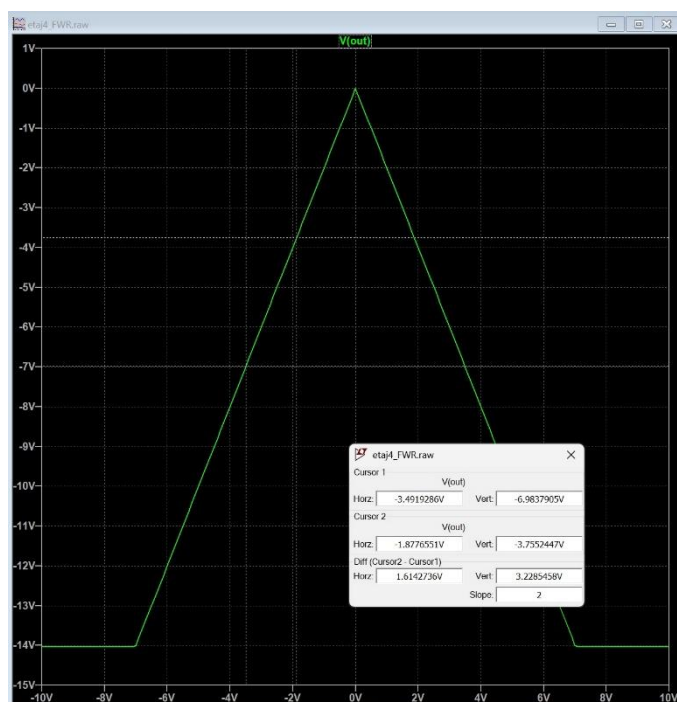
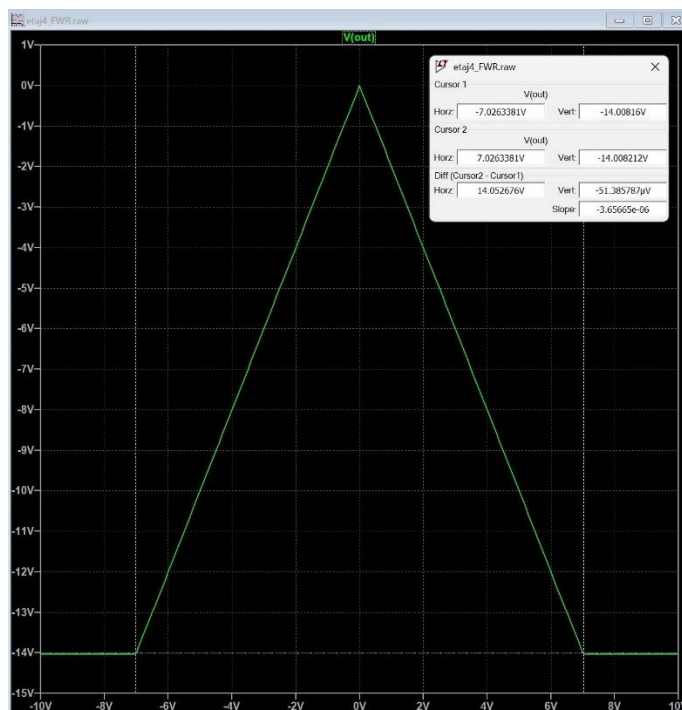


Figure (30): Gain measurement from DC analysis

3.3.7. Transient

A transient analysis was performed using a sinusoidal input signal $V_{in} = \text{SINE}(0,1 \text{ V}, 100 \text{ Hz})$ to verify the dynamic behavior of the full-wave rectifier.

The simulation confirms correct full-wave rectification, as the output signal remains unipolar while preserving the waveform shape.

From the time-domain waveforms, the peak input amplitude is approximately $V_{in,pk} \approx 1 \text{ V}$, while the peak output amplitude is $V_{out,pk} \approx 2 \text{ V}$, confirming the designed linear gain of approximately $|A_v| = 2 \text{ V/V}$. The rectified output follows the absolute value of the input signal with minimal distortion, demonstrating proper operation of the precision rectifier topology over the tested signal range.

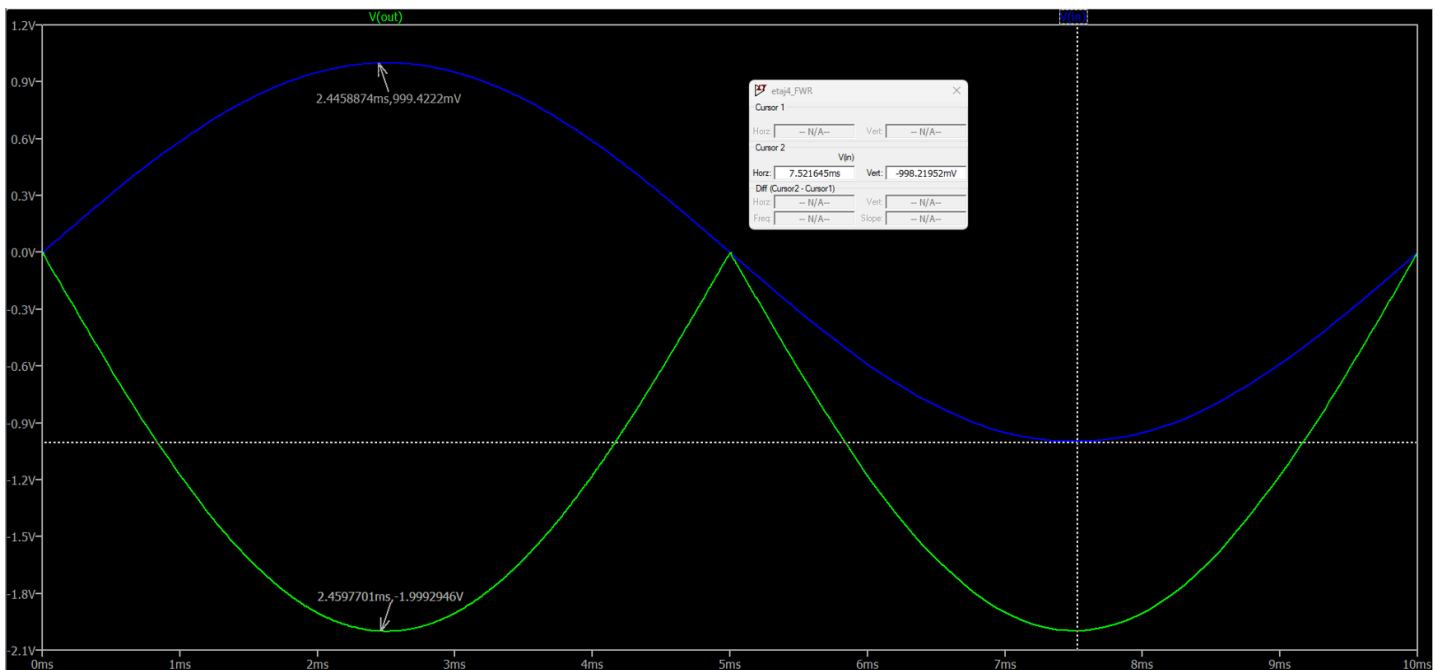


Figure (31): Time-domain verification of the precision full-wave rectifier operation (input vs. output waveforms)

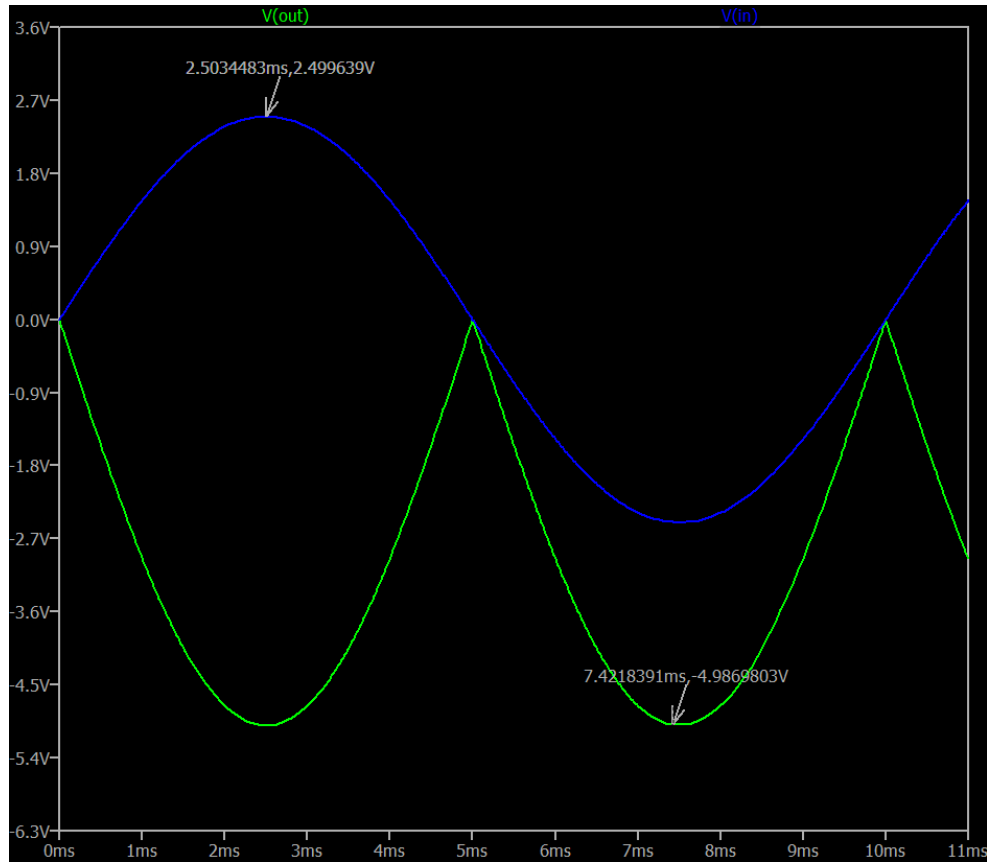


Figure (32)

The previous figure, Figure (32), shows that for the maximum input amplitude of the fourth stage, the output is not saturated. The maximum input amplitude of the full-bridge rectifier was calculated by multiplying the maximum input amplitude of the first stage with the gains of the first three stages.

$$V_{in_{4max}} = V_{in_{1max}} \cdot A_{vv_1} \cdot A_{vv_2} \cdot A_{vv_3} \cong 2.5V \quad (25)$$

Thus, the maximum amplitude of the fourth stage lies in the input voltage interval for which the output is not saturated $[-7V, 7V]$.

4. Digital Interface for Analog Signal Control

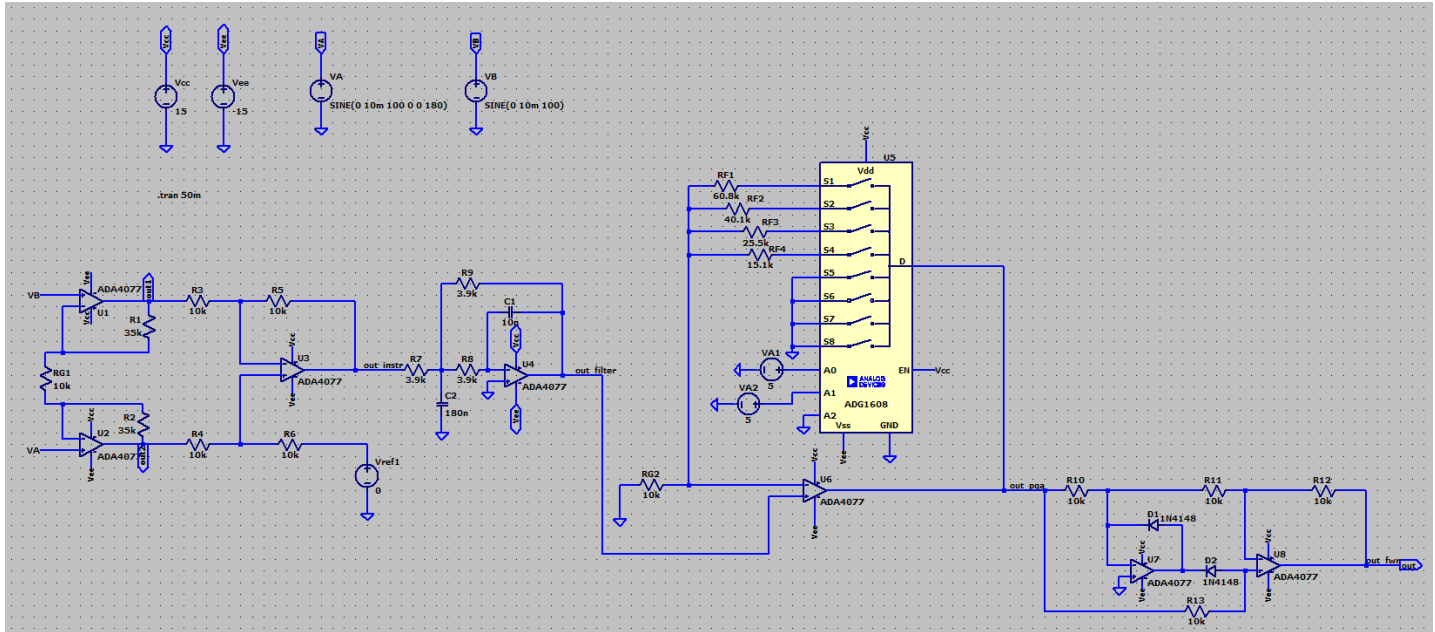


Figure (33): Full circuit

Figures (34) and (35) illustrate the transient response of the complete system, with the operation of each stage shown in separate plot panels. Two extreme configurations are analyzed: in the first case (logic A1-A0 = 1-1), the input signal has the maximum amplitude and the PGA operates at minimum gain, while in the second case (logic A1-A0 = 0-0), the input signal has the minimum amplitude and the PGA is set to its maximum gain.

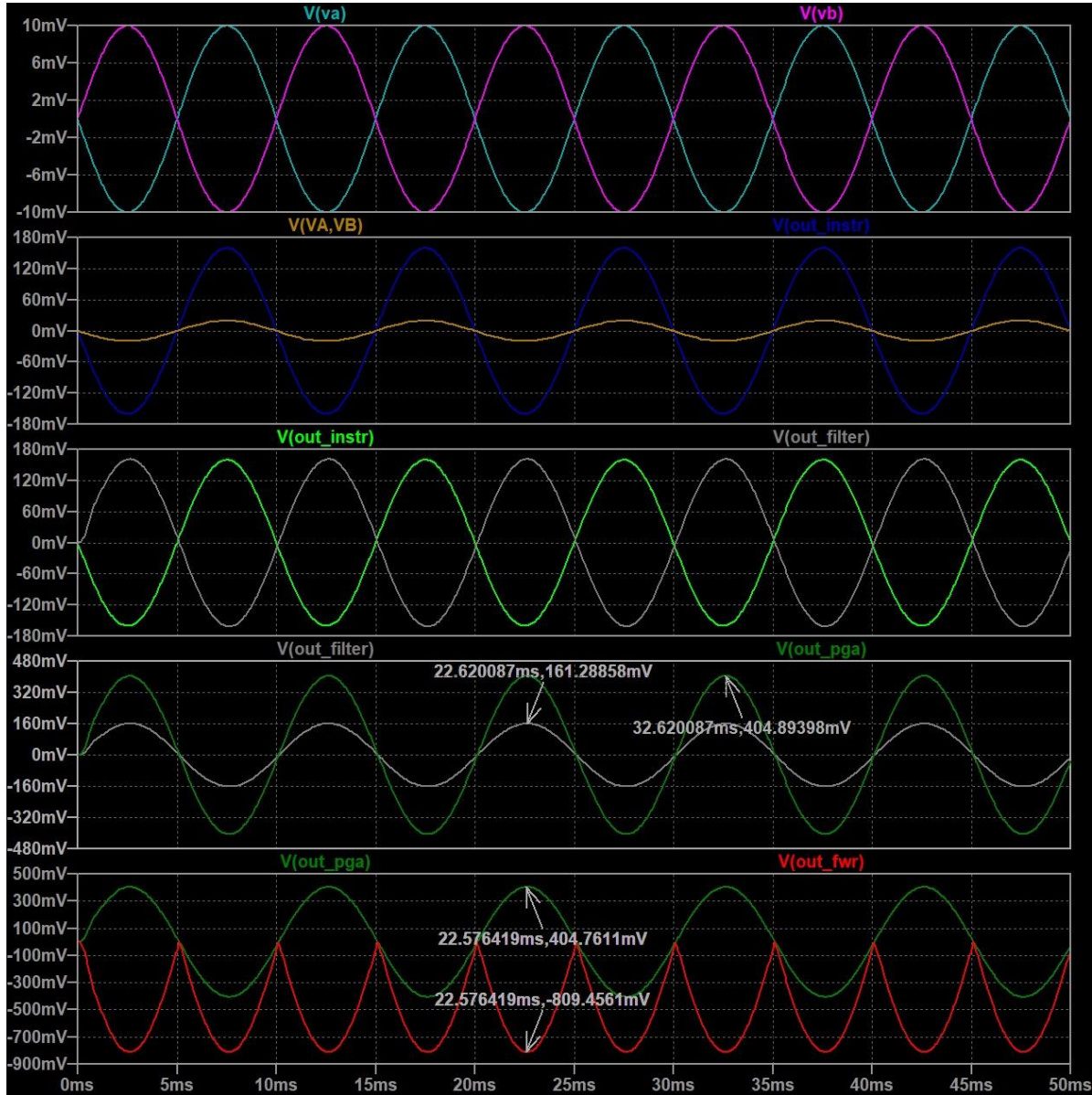


Figure (34): Time-domain response of the complete system (maximum input amplitude, minimum PGA gain $\rightarrow$ 1-1 logic)

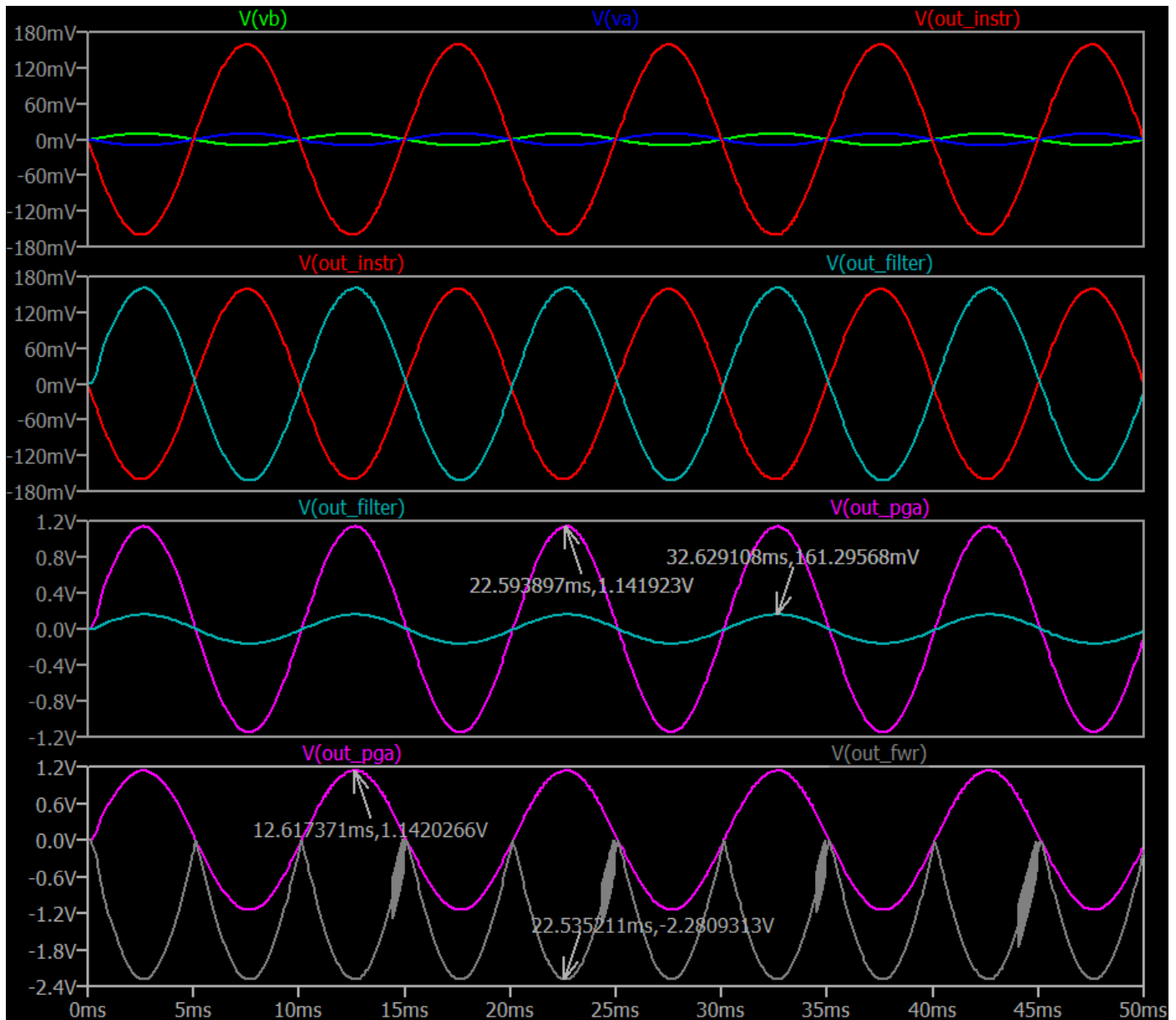


Figure (35): Time-domain response of the complete system (minimum input amplitude, maximum PGA gain
 → 0-0 logic)



5. Conclusion

The signal-conditioning chain was successfully designed, simulated, and validated against the project specifications. For all stages, the measured results closely match the theoretical values, with small deviations mainly caused by component tolerances and non-ideal op-amp behavior. The largest error appears in the gain-bandwidth product of the instrumentation amplifier, while all other parameters remain within acceptable limits, confirming that the system meets the required performance.

	Stage 1	Stage 2			Stage 3				Stage 4
	Avv [V/V]	fc [kHz]	Q	H0	Avv1 [dB]	Avv2 [dB]	Avv3 [dB]	Avv4 [dB]	H0
Measured	7.995	0.96	1.41	1.00175	7.99	10.99	13.99	16.96	1.001
Specified	8	1	1.41	1	8	11	14	17	1
Relative error [%]	0.0625	4	0	0.175	0.125	0.0909	0.0714	0.2353	0.1